

Multi-Agent Clinical Intelligence System

AI-Patient Care Assistant v2.0 • Clinical Decision Support System

Agent Architecture

Data Flow

Skills Framework

Safety & Security

Agent Architecture Flow

Orchestrator → Agents → Skills → Tools → LLM

Current runtime: the Express server calls **DocumentTypeRouter**, which classifies the PDF and routes it to a specialized extractor agent. **ReActExtractionAgent** exists in the repository but is optional rather than the default processing path. Processing Insights are served separately from `analytics.sqlite` through `/api/analytics/overview`.

LEVEL 5



Orchestrator

Routes requests • Coordinates workflows



Document Type Router

Auto-detects doc type • Routes to specialist

Classifier



Discharge Extractor

8 Skills • 30-45s

ReAct



Outpatient Extractor

OPD Records • Consultation data

ReAct



Lab Report Extractor

Lab results • Reference ranges

ReAct



Chart Note Extractor

Progress notes • SOAP format

ReAct



Doctor Assistant

7 Sub-agents • 3-8s

Multi-Agent



Chart Note Composer

1 Skill • 15-25s

ReAct



Dashboard Presenter

3 Skills • UI Components

Transform

LEVEL 4



Query Intent



Record Context



External Knowledge



Safety Guard

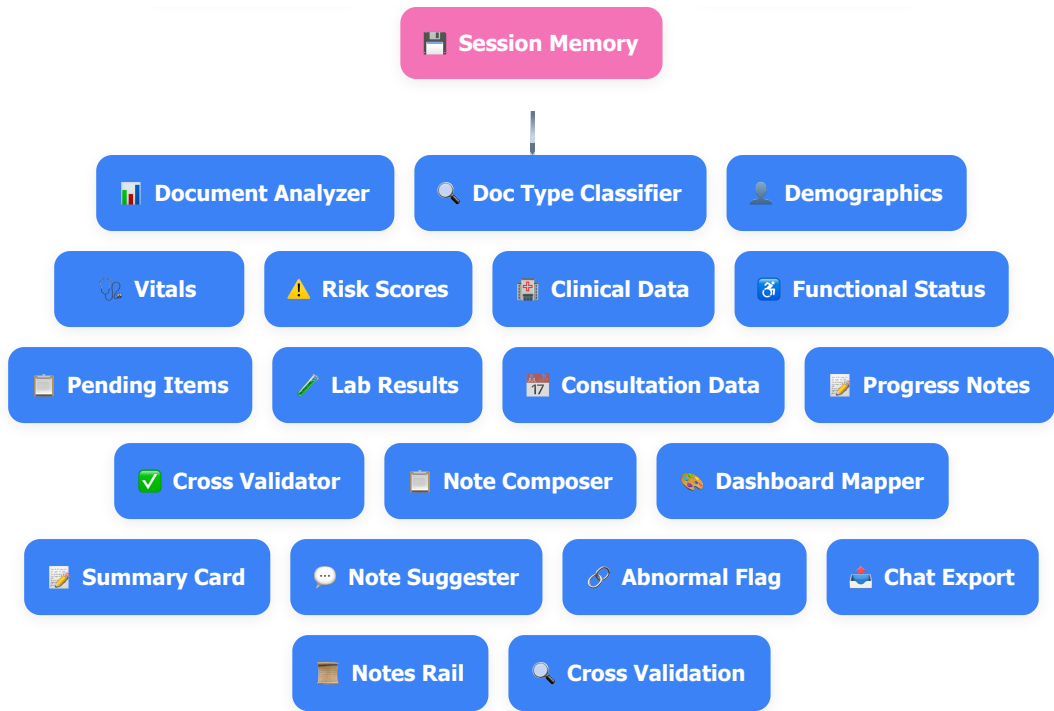


Answer Composer

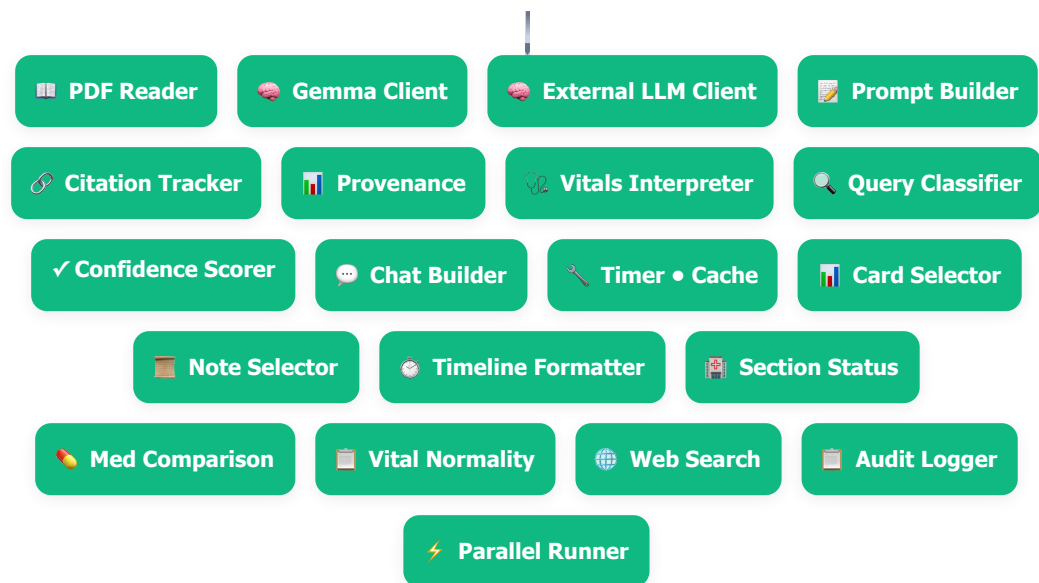


Action Router

LEVEL 3



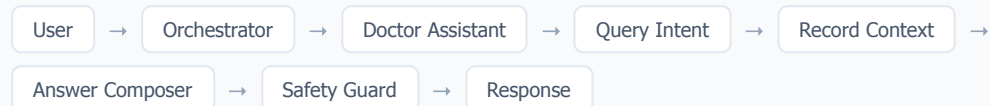
LEVEL 2



LEVEL 1



Example: "Why is BP low?"



Orchestrator (1) Primary Agents (8) • Sub-Agents (7) Skills (20) Tools (35)

LLM

Data Flow Architecture

Three operational flows, one storage boundary, and a single grounded path back to the clinician

OPERATIONS VIEW

Clinical Request to Grounded Output

Read this left to right. Each lane shows what the clinician starts, what the agents do, what gets persisted, how the output is validated, and what returns to the workspace.

Write to Storage

Read from Storage

LLM + Safety Check

ORIGIN

Clinician

Uploads records, asks chart questions, and drafts notes inside the workspace.



INTERACTION SURFACE

Clinical Workspace

One interface for patient summary, doctor chat, chart-note editing, timeline review, and citation inspection.

Patient Summary

Doctor Chat

Chart Note Editor

Timeline

Citation Viewer

LANE 1

Document Ingestion

Clinician uploads discharge, outpatient, lab, or chart-note source material.

Upload + route start



AGENT ACTION

Router + Specialist Extraction

Document type router selects the correct extractor, then extraction agents normalize content into structured fields with citations.

Router <1s • Extraction 30-45s



WRITE

Structured Clinical Store

Patient entities, vitals, labs, risk scores, citations, and audit metadata are persisted as the system record.

Single source of truth



VALIDATION

Grounding Check

Outputs are checked for structured completeness, provenance, and policy compliance before they can be shown.

No ungrounded extraction output



RESULT

Summary + Timeline Ready

The workspace renders normalized data into patient summary cards, timeline events, and evidence-linked views.

Read models for UI



LANE 2

Doctor Question

Clinician asks a patient-specific question in chat from the active workspace session.

Conversation start



AGENT ACTION

Doctor Assistant Mesh

Intent, record context, safety, answer composition, and session memory collaborate to prepare a grounded response request.

Chat 3-8s

READ + WRITE

Context, Session, Citations

The assistant reads prior extractions and chat state, then writes response state, citations, and audit entries back to storage.

Session memory stays persisted



LLM PATH

Gemma + Safety Validation

Self-hosted inference generates the answer, then refusal policy, confidence checks, and source grounding validate it before return.

PHI remains on-premise



RESULT

Cited Assistant Reply

The workspace shows a grounded answer with source-backed citations instead of a free-floating LLM response.

Evidence visible to clinician

LANE 3

Chart-Note Drafting

Clinician opens a SOAP workflow to generate or refine a chart note from current patient context.

Draft + review start



AGENT ACTION

Chart Note Composer

The composer runs section-by-section drafting for subjective, objective, assessment, and plan with review checkpoints.

SOAP 15-25s



WRITE

Drafts, Versions, Provenance

Every note revision is versioned with supporting citations, timestamps, and actor attribution for downstream auditability.

Version history retained



VALIDATION

Review Before Finalization

Generated note content is checked for grounding and safety before it is surfaced as a candidate clinical draft.

Human-in-the-loop required



RESULT

Final Note in Workspace

The clinician receives a reviewable draft with lineage intact, rather than an opaque final document.

Review, edit, approve

STORAGE DISCIPLINE

All writes pass through the same storage boundary. Extraction output, chat state, note drafts, citations, and audit events are persisted in one place.

READ DISCIPLINE

The UI does not bypass storage. Summary panels, chat context, and timelines are all hydrated from stored read models rather than ad hoc state.

INFERENCE GUARDRAIL

Gemma is self-hosted. PHI stays on-premise, and every response path is expected to meet grounding and validation checks before display.

OPERATIONAL ENVELOPE

Router <1s · Chat 3-8s · SOAP 15-25s · Extraction 30-45s

Skills Framework

20 Reusable AI Capabilities organized by category

Document Classification (1)

Document Type Classifier

Auto-detect doc type

Extraction Skills (10)

Document Analyzer

PDF · Structure

Demographics Extractor	Patient Info
Risk Scores Extractor	Fall • DVT • EWS
Vitals Extractor	BP • Pulse • SpO2
Functional Status Extractor	Mobility • Assistance
Clinical Data Extractor	Dx • Meds • Labs
Pending Items Extractor	LLM-Only • Labs • Radiology
Lab Results Extractor	CBC • Chemistry • Ref ranges
Consultation Data Extractor	OPD • Clinic notes
Progress Notes Extractor	SOAP • Resident notes
Cross Validator	Source Check • Citations

✓ Validation Skills (2)

Cross Validator	Verify • Citations
Cross Validation Agent	Confidence • Scoring

📄 Generation Skills (1)

Chart Note Composer	SOAP Assembly
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🗺️ Presentation Skills (3)

Dashboard Mapper	JSON → UI
Summary Card Builder	Card Format
Notes Rail Builder	Timeline Notes

💬 Chat Skills (3)

Note Update Suggester	Draft Updates
Abnormal Flag Action	Decision Support
Chat Export Builder	PDF Appendix

Tools Layer (35 utilities)

DOCUMENT ROUTER (1)

📁 Type Classifier

PDF (1)

📖 PDF Reader

LLM (4)

🧠 Gemma Client

🧠 External LLM Client

📄 Prompt Builder

🔗 Citation Tracker

CLINICAL (2)

🩺 Vitals Interpreter

🏠 Provenance Builder

ORCHESTRATION (2)

 Audit Logger

 Parallel Runner

PRESENTATION (4)

 Card Selector

 Section Status

 Note Selector

 Timeline Formatter

CHAT (20)

 Query Classifier

 Section Hint Resolver

 Citation Assembler

 Medical Web Search

 Confidence Scorer

 Chat Prompt Builder

 Vital Normality

 Medication Comparison

 Section Context Fetch

 Record Context Search

 Refusal Policy

 Provenance Gate

 Source Policy

 Source Router

 Source Health

 External Source Ranker

 External Citation Normalizer

 External Query Planner

 Drug Entity Resolver

 Drug Fact Extractor

Tools are low-level utilities used by skills. Skills compose tools into reusable capabilities.

Safety & Security Architecture

Multi-layer protection for clinical AI systems

Hallucination Detection

Multi-Layer Validation:

1. Source Verification

Every fact checked against source PDF

2. Citation Requirements

Character-level citations mandatory

3. Confidence Scoring

Low confidence = blocked output

4. Refusal Policy

Unsafe responses automatically refused

Data Privacy & PHI Protection

PHI Protection Measures:

● LLM self-hosted on-premises (no external API calls)

● PHI encrypted at rest

● Audit trail for all data access

● Session isolation per document

● External search requires consent

Safety Guard Agent

Validates all outputs before returning to user:

Factual Accuracy

Cross-check with sources

Confidence Level

Score threshold check

Clinical Safety

No harmful advice

Source Attribution

Citations attached

⚠️ Low confidence or hallucination detected = Output blocked, user notified

External Search Consent Flow

User asks question requiring external medical knowledge:

1. Classify

→

2. Prompt

→

3. Execute

→

4. Response

Explicit consent required before any external medical source search

Approved sources only • Patient data never sent externally

Error Handling & Fallbacks

LLM Errors

→ Retry with fallback

Validation Failures

→ Return safe message

Search Down

→ Use internal only

Timeout

→ Graceful degradation

All errors logged • Critical errors trigger alerts • Helpful messages to users

Audit Trail & Provenance

Every action is tracked:

● User queries with timestamps

● LLM prompts + responses

● Safety checks + refusals

● Agent decisions + reasoning

● Data reads/writes provenance

● External consent events

Security Principles

• Never trust LLM output blindly • Always verify against source • Patient safety first •

• PHI stays on-premises • External search requires consent • All actions auditable • Human-in-the-loop for clinical decisions

• Fail-safe defaults • Graceful degradation • Clear error messages • No raw exceptions to users

Safety Statistics

100%
Outputs Validated

Char-level
Citation Precision

0
External API Leaks

Always
Consent Required

24/7
Audit Trail Active

On-Prem
LLM Deployment